Repeat Mid-term, Spring Semester, 2023
CO 315 (B.Tech), CSE, Tezpur University,
FM: 40, FT: 120 mins

1. Define the type of the following destination addresses: 3
   a. 4A:30:10:21:10:1A
   b. 47:20:1B:2E:08:EE
   c. FF:FF:FF:FF:FF:FF

2. What are the functions of EtherType and Protocol Type in the headers? 2

3. Write down any three standard Ethernet implementations that use coaxial, UTP and fiber respectively as the cables. Denote them in terms of X BandType Y where X and Y are datarate and cable type. Also write the topologies supported by each of them. 3+3+3

4. An application generates IP packet of 5000 bytes minimum. At the destination routers, internal network supports only Ethernet. Fragmentation is required perhaps. Show the fragments offsets and fragmentation flags of each fragment. 7

5. What is the significance of delay-bandwidth product in MAC? Deduce a relation between MAC efficiency and one-way delay-bandwidth product. 1+2

6. Write down the differences among the protocol stacks: IEEE 802.3, OSI and TCP/IP. 3

7. Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the application layer, network layer and the data link layer during a transmission from S to D. Justify and explain properly. 3

8. How does the NIC connect to memory and network? What are the types of byte ordering? What is the byte-order of the network bytes? Show how the address 47:20:1B:2E:08:EE is sent out on line with direction of flow. 2+2+1+2

9. Differentiate among 1-P, non-P and p-P CSMAs. 3